Helix Fixed Workable Items

Generated from docs/workable_items.db. This document does not assert remote HelixTrack synchronization.

[ATM-001] Establish core document, cache, settings, and synchronization primitives

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** High **Reopens:** 0

Description: Establish core document, cache, settings, and synchronization primitives. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Core Go module and shared domain model.

Reproduction / current observation: Core tests, race detector, vet, coverage, and benchmarks execute against the current tree.

Acceptance: Shared primitives remain deterministic, race-free, and covered by repeatable tests.

Closure criteria: Shared primitives remain deterministic, race-free, and covered by repeatable tests.

Evidence: evidence/core/final-go-test.log; evidence/core/final-go-race.log; evidence/core/final-go-vet.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-002] Implement multi-format detection and bounded document parsing

Status: Implemented (→ Fixed.md) **Type:** Feature **Severity:** High **Reopens:** 0

Description: Implement multi-format detection and bounded document parsing. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: EPUB, CBZ, FB2, MOBI, TXT, and related registry paths.

Reproduction / current observation: Format fixtures and seven fuzz targets exercise malformed and valid inputs without claiming full rendering or conversion.

Acceptance: Detection and parsing stay bounded, deterministic, and explicit about unsupported fulfillment stages.

Closure criteria: Detection and parsing stay bounded, deterministic, and explicit about unsupported fulfillment stages.

Evidence: evidence/core/final-go-test.log; evidence/core/final-go-fuzz-formats.log; docs/formats.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-003] Canonicalize ISBN identity across equivalent representations

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Critical **Reopens:** 0

Description: Canonicalize ISBN identity across equivalent representations. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Work, edition, file identification and ingest reconciliation.

Reproduction / current observation: ISBN-10/13, separators, schemes, and identifier ordering previously risked divergent identity.

Acceptance: Equivalent valid ISBNs converge while invalid identifiers fall back to content identity.

Closure criteria: Equivalent valid ISBNs converge while invalid identifiers fall back to content identity.

Evidence: core/isbn/isbn.go; server/internal/ingest/isbn_identity_test.go; evidence/server/final-go-test.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-004] Index local and remote libraries in place without reorganizing source bytes

Status: Implemented (→ Fixed.md) **Type:** Feature **Severity:** Critical **Reopens:** 0

Description: Index local and remote libraries in place without reorganizing source bytes. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Local, HTTP, WebDAV, OPDS, FTP, SFTP, SMB, S3, and rclone connector surfaces.

Reproduction / current observation: Connectors preserve source paths and capabilities; deployment-specific provider validation remains separate open work.

Acceptance: Scans index metadata and references without Calibre-style copying or destructive relocation.

Closure criteria: Scans index metadata and references without Calibre-style copying or destructive relocation.

Evidence: server/internal/sources; docs/sources.md; evidence/server/final-go-test.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-005] Make OPDS acquisition identity collision-safe and atomically replaceable

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Critical **Reopens:** 0

Description: Make OPDS acquisition identity collision-safe and atomically replaceable. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: OPDS feed pagination and acquisition-map generations.

Reproduction / current observation: Credential-free public keys, deterministic collision fragments, previous-generation reuse, and stale pruning are exercised.

Acceptance: Concurrent feed refreshes cannot expose partially replaced acquisition maps or unstable colliding keys.

Closure criteria: Concurrent feed refreshes cannot expose partially replaced acquisition maps or unstable colliding keys.

Evidence: server/internal/sources/opds.go; server/internal/sources/opds_test.go; evidence/server/final-go-test.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-006] Revalidate content before strong ETag and byte-range responses

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Critical **Reopens:** 0

Description: Revalidate content before strong ETag and byte-range responses. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Local and remote content GET and HEAD fulfillment.

Reproduction / current observation: Live size and modification-time checks gate scan-time hashes; local GET uses one revalidated descriptor.

Acceptance: Changed or deleted bytes never retain a stale strong validator or splice old and new ranges.

Closure criteria: Changed or deleted bytes never retain a stale strong validator or splice old and new ranges.

Evidence: server/internal/sources/local.go; server/internal/httpapi/content_test.go; evidence/server/final-go-test.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-007] Preserve persistent HTTP connections during identity compression fallback

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Critical **Reopens:** 0

Description: Preserve persistent HTTP connections during identity compression fallback. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Brotli and gzip middleware on HTTP/1.1 persistent connections.

Reproduction / current observation: An unused negotiated compressor was finalized after an identity attachment, appending an empty stream and resetting the next response.

Acceptance: Identity fallback disables compressor finalization; raw-socket Brotli and gzip keepalive regressions and mutation reject the defect.

Closure criteria: Identity fallback disables compressor finalization; raw-socket Brotli and gzip keepalive regressions and mutation reject the defect.

Evidence: evidence/live-server/post-fix-repeat-01/results.json; evidence/live-server/post-fix-daemon-restart/results.json; evidence/go-mutation/results.json

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-008] Provide HTTP/1.1, HTTP/2, HTTP/3, TLS, and content negotiation transports

Status: Implemented (→ Fixed.md) **Type:** Feature **Severity:** High **Reopens:** 0

Description: Provide HTTP/1.1, HTTP/2, HTTP/3, TLS, and content negotiation transports. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Gin server transport and live protocol verification.

Reproduction / current observation: Current-tree binaries are probed over supported protocols with safe fallback and explicit TLS floors.

Acceptance: Transport evidence is regenerated from the current tree and distinguishes unavailable protocol support.

Closure criteria: Transport evidence is regenerated from the current tree and distinguishes unavailable protocol support.

Evidence: evidence/live-transport/final-current-tree/results.json; docs/deployment.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-009] Reconcile durable PostgreSQL catalogs atomically after source scans

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Critical **Reopens:** 0

Description: Reconcile durable PostgreSQL catalogs atomically after source scans. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: PostgreSQL work, edition, file, and provenance persistence.

Reproduction / current observation: Shared validation and identity locking prevent partial reconciliation and cross-work file ownership conflicts.

Acceptance: Deleted source files and rehomed identities commit atomically with matching memory-store semantics.

Closure criteria: Deleted source files and rehomed identities commit atomically with matching memory-store semantics.

Evidence: server/internal/catalog/postgres.go; evidence/server/final-go-live-services.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-010] Implement monotonic KOReader progress synchronization with Redis

Status: Implemented (→ Fixed.md) **Type:** Feature **Severity:** High **Reopens:** 0

Description: Implement monotonic KOReader progress synchronization with Redis. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: KOSync account, document progress, and Redis optimistic transaction paths.

Reproduction / current observation: WATCH/MULTI/EXEC and monotonic progress behavior are exercised against a real Redis service.

Acceptance: Progress cannot regress under stale updates and live-service evidence remains separate from unit doubles.

Closure criteria: Progress cannot regress under stale updates and live-service evidence remains separate from unit doubles.

Evidence: server/internal/kosync; evidence/server/final-go-live-services.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-011] Deliver an honest responsive web catalog prototype

Status: Implemented (→ Fixed.md) **Type:** Feature **Severity:** Medium **Reopens:** 0

Description: Deliver an honest responsive web catalog prototype. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: React catalog-management demonstration across four viewport classes.

Reproduction / current observation: The interface discloses sample-only data and the absence of a backend connection.

Acceptance: Lint, production build, and four-viewport browser checks pass without presenting sample data as live catalog state.

Closure criteria: Lint, production build, and four-viewport browser checks pass without presenting sample data as live catalog state.

Evidence: evidence/ui/final-npm-build.log; evidence/ui/final-run.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-012] Establish repeatable deterministic core and server qualification

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** Critical **Reopens:** 0

Description: Establish repeatable deterministic core and server qualification. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Unit, integration, race, vet, repeat, and coverage campaigns.

Reproduction / current observation: Current-tree logs include genuine failures alongside passing reruns; no result is inferred from command invocation alone.

Acceptance: Every published result links machine output and distinguishes pass, fail, skip, and not-run states.

Closure criteria: Every published result links machine output and distinguishes pass, fail, skip, and not-run states.

Evidence: evidence/core; evidence/server; docs/testing/results

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-013] Fuzz all exposed format and byte-range parsers

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** High **Reopens:** 0

Description: Fuzz all exposed format and byte-range parsers. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Formats, CBZ, EPUB, FB2, MOBI, TXT, and single-range parsing.

Reproduction / current observation: Seven bounded fuzz campaigns ran against the current tree with retained logs.

Acceptance: Fuzz targets terminate without crashes or unbounded parser behavior in the recorded campaigns.

Closure criteria: Fuzz targets terminate without crashes or unbounded parser behavior in the recorded campaigns.

Evidence: evidence/core/final-go-fuzz-formats.log; evidence/core/final-go-fuzz-cbz.log; evidence/core/final-go-fuzz-epub.log; evidence/core/final-go-fuzz-fb2.log; evidence/core/final-go-fuzz-mobi.log; evidence/core/final-go-fuzz-txt.log; evidence/server/final-go-fuzz-single-range.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-014] Prove regression tests reject targeted restoration of fixed defects

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** High **Reopens:** 0

Description: Prove regression tests reject targeted restoration of fixed defects. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Go mutation campaign across core identity, ingest, sources, and HTTP behavior.

Reproduction / current observation: Twenty-one targeted mutants were rejected and original-file hashes remained unchanged.

Acceptance: Every mutant fails its focused test and the original tree remains byte-identical after restoration.

Closure criteria: Every mutant fails its focused test and the original tree remains byte-identical after restoration.

Evidence: evidence/go-mutation/results.json; evidence/go-mutation/final-verifier.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-015] Exercise real PostgreSQL, pgvector, and Redis integrations

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** Critical **Reopens:** 0

Description: Exercise real PostgreSQL, pgvector, and Redis integrations. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Durable catalog, vector memory, KOSync, and internal progress services.

Reproduction / current observation: Live-service tests ran separately from in-memory and miniredis tests.

Acceptance: Integration claims require real service output and preserve unavailable credentials or services as non-pass.

Closure criteria: Integration claims require real service output and preserve unavailable credentials or services as non-pass.

Evidence: evidence/server/final-go-live-services.log

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-016] Verify backup and restore recovery against the current tree

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** High **Reopens:** 0

Description: Verify backup and restore recovery against the current tree. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Catalog and synchronization backup/restore workflow.

Reproduction / current observation: The current-tree recovery verifier completed thirteen checks before the latest compression edit and requires final refresh.

Acceptance: A clean recovery reproduces catalog and sync state without secret disclosure.

Closure criteria: A clean recovery reproduces catalog and sync state without secret disclosure.

Evidence: evidence/backup-restore/final-current-tree/results.json

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-017] Prevent credentials and sensitive values from entering source and evidence

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** Critical **Reopens:** 0

Description: Prevent credentials and sensitive values from entering source and evidence. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Constitution, project instructions, connectors, diagnostics, and release scans.

Reproduction / current observation: Secret-handling constraints are mandatory and endpoint credentials are rejected; final archive scanning remains a separate release gate.

Acceptance: No credential is committed, printed, logged, or included in evidence or archives.

Closure criteria: No credential is committed, printed, logged, or included in evidence or archives.

Evidence: AGENTS.md; CLAUDE.md; spec/00_constitution.md; server/internal/sources/http.go

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-018] Keep deterministic metadata evidence authoritative over AI proposals

Status: Implemented (→ Fixed.md) **Type:** Feature **Severity:** High **Reopens:** 0

Description: Keep deterministic metadata evidence authoritative over AI proposals. This registry entry records the implemented or remaining work without expanding evidence beyond its actual scope.

Affected scope: Identification pipeline, provenance, confidence, and AI memory.

Reproduction / current observation: Embedded deterministic identifiers take precedence and provider proposals retain confidence and provenance.

Acceptance: Low-confidence AI output is never silently promoted to verified metadata fact.

Closure criteria: Low-confidence AI output is never silently promoted to verified metadata fact.

Evidence: server/internal/identify; server/internal/ingest; docs/architecture.md

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-028] Operate the SQLite workable-items single source of truth and synchronized exports

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** Critical **Reopens:** 0

Description: Operate the authoritative SQLite registry with complete lifecycle commands, evidence-linked history, deterministic document exports, drift detection, disaster recovery, and an explicitly offline HelixTrack outbox.

Affected scope: Workable-item database, lifecycle history, exports, validation, and drift gates.

Reproduction / current observation: Twelve real-SQLite tests, ten concurrent writers, one thousand stress rows, SIGKILL recovery, corrupt-database recovery, and four paired mutation gates passed; remote HelixTrack delivery remains a separate open item.

Acceptance: The database validates, all derived documents regenerate without drift, round trips preserve content, lifecycle and stress gates pass, and no remote push is claimed without a receipt.

Closure criteria: The database validates, all derived documents regenerate without drift, round trips preserve content, lifecycle and stress gates pass, and no remote push is claimed without a receipt.

Evidence: evidence/workable-items/results.json; evidence/workable-items/final-verifier.log; evidence/workable-items/pdf-qa

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.

[ATM-030] Publish all current test results and limitations under docs/testing/results

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** High **Reopens:** 0

Description: Publish a unified machine-readable and Markdown result index covering current passes, preserved failures, partial research, unavailable runtimes, unexecuted external checks, and withheld final archives.

Affected scope: Machine-result index for unit, live, UI, integration, deployment, and non-run checks.

Reproduction / current observation: The generated index contains 27 scoped outcomes with evidence existence and SHA-256 metadata; historical failures and non-run targets remain visible and release qualification is NOT READY.

Acceptance: Every registered suite and unavailable target links its current artifact or explicit non-run statement, and publication never converts missing scope into a pass.

Closure criteria: Every registered suite and unavailable target links its current artifact or explicit non-run statement, and publication never converts missing scope into a pass.

Evidence: docs/testing/results/summary.json; docs/testing/results/index.md; scripts/publish_test_results.py

Forensic anchor: Retrospective evidence-backed baseline imported on 2026-10-29; registry timestamps are import times, not claimed original implementation dates.